2 Real Analysis 2023
Tutorial 9

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The topology of $\mathbb{R}$

Problem 34 Decide whether the following sets are open, closed, or neither. Give a justification.

(a) $\mathbb{N}$,

(b) $\mathbb{Q}$,

(c) $B = \{x \in \mathbb{R} : x \neq 5\}$,

(d) $C = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$,

(e) $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

Solution.

(a) The set $\mathbb{N}$ is closed. This is because any convergent sequence in $\mathbb{N}$ is eventually constant, and thus its limit is in $\mathbb{N}$. The set $\mathbb{N}$ is not open. For $n \in \mathbb{N}$, any interval of the form $(n - \varepsilon, n + \varepsilon)$, $\varepsilon > 0$, contains points from $\mathbb{R} \setminus \mathbb{N}$.

(b) The set $\mathbb{Q}$ is not closed. Indeed, the set of all limit points of $\mathbb{Q}$ is $\mathbb{R}$, since for any $x \in \mathbb{R}$ we can find a sequence in $\mathbb{Q} \setminus \{x\}$ that converges to x. The set $\mathbb{Q}$ is not open. For $r \in \mathbb{Q}$, any interval of the form $(r - \varepsilon, r + \varepsilon)$, $\varepsilon > 0$, contains points from $\mathbb{R} \setminus \mathbb{Q}$.

(c) $B = (-\infty, 5) \cup (5, \infty)$ is a union of two open sets and hence open. B is not closed because B does not contain its limit point 5 (for example, the sequence $\left(5 + \frac{1}{n}\right)_{n=1}^{\infty}$ in B converges to 5).

(d) $C = [-2, -1] \cup [1, 2]$ is a union of two closed sets and hence closed. C is not open. For example, $2 \in C$, but any interval $(2 - \varepsilon, 2 + \varepsilon)$, $\varepsilon > 0$, contains points from $\mathbb{R} \setminus C$.

(e) The set D is not open. For example, $1 \in D$, but any interval $(1 - \varepsilon, 1 + \varepsilon)$, $\varepsilon > 0$, contains points from $\mathbb{R} \setminus D$. The set D is not closed, since it does not contain its limit point 0 (0 is a limit point of D because $\lim_{n\to\infty} \frac{1}{n} = 0$).

Problem 35 Consider the set $S = \{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup [3, 4) \cup \{5\}$. Find

(a) all its limit points,

(b) all its isolated points,

(c) its closure $\overline{S}$.

Solution.

(a) The set of all limit points of S is $S' = \{0\} \cup [3,4]$. Indeed, if $x < 0$, then any sequence convergent to x must be eventually in $(-\infty, 0)$, but there are no points of $S \setminus \{x\}$ in $(-\infty, 0)$, and thus x is not a limit point of S. The point 0 is a limit point of S because $\lim_{n\to\infty} \frac{1}{n} = 0$. If $x \in (0,3)$, then we can always find a small interval of the form $(x-\varepsilon, x+\varepsilon)$, $\varepsilon > 0$, that does not contain points from $S \setminus \{x\}$, and thus x is not a limit point of S. Any point $x \in [3,4)$ is a limit point of S, because the sequence $\left(x + \frac{1}{n}\right)_{n=n_0}^{\infty}$ for n_0 large enough lies in $S \setminus \{x\}$ and converges to x. Similarly, $x = 4$ is a limit point of S, because the sequence $\left(4 - \frac{1}{n}\right)_{n=1}^{\infty}$ lies in $S \setminus \{4\}$ and converges to 4. Finally, if $x \in (4, \infty)$, then any sequence convergent to x must be eventually in $(4, \infty)$, but there are no points of $S \setminus \{x\}$ in $(4,\infty)$, and thus x is not a limit point of S.

(b) Recall that x is an isolated point of S if $x \in S$ and x is not a limit point of S. Using the result of (a), we conclude that the test of isolated points of S is $\left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{5\}$.

(c) $\overline{S} = S \cup S' = \{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup [3,4] \cup \{5\}$.

Problem 36 Let $A, B \subseteq \mathbb{R}$. Prove the following properties of the closures.

(a) $A \subseteq B \implies \overline{A} \subseteq \overline{B}$,

(b) $\overline{\overline{A}} = \overline{A}$,

(c) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution.

(a) Recall that $\overline{A}$ is the smallest closed set containing A. If $A \subseteq B$, then $\overline{B}$ is a closed set and $A \subseteq B \subseteq \overline{B}$. It follows that $\overline{A} \subseteq \overline{B}$.

(b) $\overline{\overline{A}}$ is a closed set containing $\overline{A}$, i.e. $\overline{A} \subseteq \overline{\overline{A}}$. On the other hand, $\overline{A}$ is a closed set containing $\overline{A}$, and $\overline{\overline{A}}$ is the smallest closed set with this property. Thus, $\overline{\overline{A}} \subseteq \overline{A}$. The two inclusions together imply $\overline{\overline{A}} = \overline{A}$.

(c) Since the union of two closed sets is closed, $\overline{A} \cup \overline{B}$ is a closed set. Since $A \subseteq \overline{A}$ and $B \subseteq \overline{B}$, we have that $A \cup B \subseteq \overline{A} \cup \overline{B}$. It follows that $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$.

On the other hand, since $A \subseteq A \cup B$, by part (a) we have $\overline{A} \subseteq \overline{A \cup B}$. Similarly, $\overline{B} \subseteq \overline{A \cup B}$. This implies $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$. The two inclusions together give $\overline{A} \cup \overline{B} = \overline{A \cup B}$.

Problem 37 Let $A \subseteq \mathbb{R}$. A point $x \in \mathbb{R}$ is called an *adherent point* of A, if any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of A.

(a) Prove that the closure $\overline{A}$ of the set A is exactly the set of all adherent points of A.

(b) Prove that

$$\begin{aligned}&\{a \in \mathbb{R} : a \text{ is an adherent point of } A\}\\ &= \{a \in \mathbb{R} : a \text{ is a limit point of } A\} \cup \{a \in \mathbb{R} : a \text{ is an isolated point of } A\}.\end{aligned}$$

Solution.

(a) Recall that $\overline{A} = A \cup A'$. If $x \in A$, then any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains the point $x \in A$. If $x \in A'$, then any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of A (different from x) by the definition of a limit point. In both cases any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of A. Thus, if $x \in \overline{A} = A \cup A'$, then x is an adherent point of A.

For the other direction, assume that x is an adherent point of A, i.e. any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of A. If $x \in A$, then also $x \in \overline{A}$. If $x \notin A$, then it is a limit point of A, and thus again $x \in \overline{A}$. Thus, if x is an adherent point of A, then $x \in \overline{A}$.

(b) By (a), the set of all adherent points of A coincides with $\overline{A} = A \cup A'$. Any point in $\overline{A}$ that is not in A' must lie in A, and thus it is an isolated point of A. Thus,

$$\overline{A} = \{a \in \mathbb{R} : a \text{ is a limit point of } A\} \cup \{a \in \mathbb{R} : a \text{ is an isolated point of } A\},$$

and the desired statement follows.

Problem 38 Let $A \subseteq \mathbb{R}$. A point $x \in \mathbb{R}$ is called an *boundary point* of A, if any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of A and a point of its complement $\mathbb{R} \setminus A$. The set δA of all boundary points of A is called the *boundary* of A.

(a) Prove that $\delta A = \delta(\mathbb{R} \setminus A)$.

(b) Determine the boundaries of the following sets: $B = \{x \in \mathbb{R} : x \neq 5\}$, $C = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$, $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, $M = (0,1] \cup \{2\} \cup [3,4]$.

Solution.

(a) We use the fact that $\mathbb{R} \setminus (\mathbb{R} \setminus A) = A$. Now x is a boundary point of A if and only if any ε-neighborhood $(x-\varepsilon, x+\varepsilon)$ contains a point of $A = \mathbb{R} \setminus (\mathbb{R} \setminus A)$ and a point of its complement $\mathbb{R} \setminus A$ if and only if x is a boundary point of $\mathbb{R} \setminus A$.

(b) $\delta B = \{5\}$. Notice that $\mathbb{R} \setminus B = \{5\}$. Any ε-neighborhood $(5-\varepsilon, 5+\varepsilon)$ contains the point $5 \in \mathbb{R} \setminus B$ as well as points from B. If $x \neq 5$, then $x \in (-\infty, 5) \cup (5, \infty)$ which is an open set, and therefore there exists an ε-neighborhood $(x-\varepsilon, x+\varepsilon) \subseteq (-\infty, 5) \cup (5, \infty) = B$. We conclude that $x \notin \delta B$.

Notice that $C = [-2, -1] \cup [1, 2]$. We have $\delta C = \{-2, -1, 1, 2\}$. Indeed, for each of this points any ε-neighborhood contains points from C and from $\mathbb{R} \setminus C$. On the other hand, if $x \in (-2, -1)$ or $x \in (1, 2)$, which are open sets contained in C, then there is an ε-neighborhood of x that contains only points from C. And if $x \in (-\infty, -2) \cup (-1, 1) \cup (2, \infty)$ which is an open set contained in $\mathbb{R} \setminus C$, then there is an ε-neighborhood of x that contains only points from $\mathbb{R} \setminus C$.

$\delta D = \{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. If $x = \frac{1}{n}$ with $n \in \mathbb{N}$, then any ε-neighborhood of $\frac{1}{n}$ contains the point $\frac{1}{n} \in D$ as well as points from $\mathbb{R} \setminus D$. Since $\lim_{n\to\infty} \frac{1}{n} = 0$, by the definition of the limit any ε-neighborhood of 0 contains points from D; but it also contains points from $\mathbb{R} \setminus D$. If $x \neq 0$ and $x \neq \frac{1}{n}$, then x lies in the open set $\mathbb{R} \setminus \left(\{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right)$, and there is a ε-neighborhood of x that contains only points from $\mathbb{R} \setminus D$.

$\delta M = \{0, 1, 2, 3, 4\}$. For each of this points any ε-neighborhood contains points from M and from $\mathbb{R} \setminus M$. On the other hand, if $x \in (0, 1) \cup (3, 4)$ which is an open set contained in M, then there is an ε-neighborhood of x that contains only points from M. And if $x \in (-\infty, 0) \cup (1, 2) \cup (2, 3) \cup (4, \infty)$ which is an open set contained in $\mathbb{R} \setminus M$, then there is an ε-neighborhood of x that contains only points from $\mathbb{R} \setminus M$.